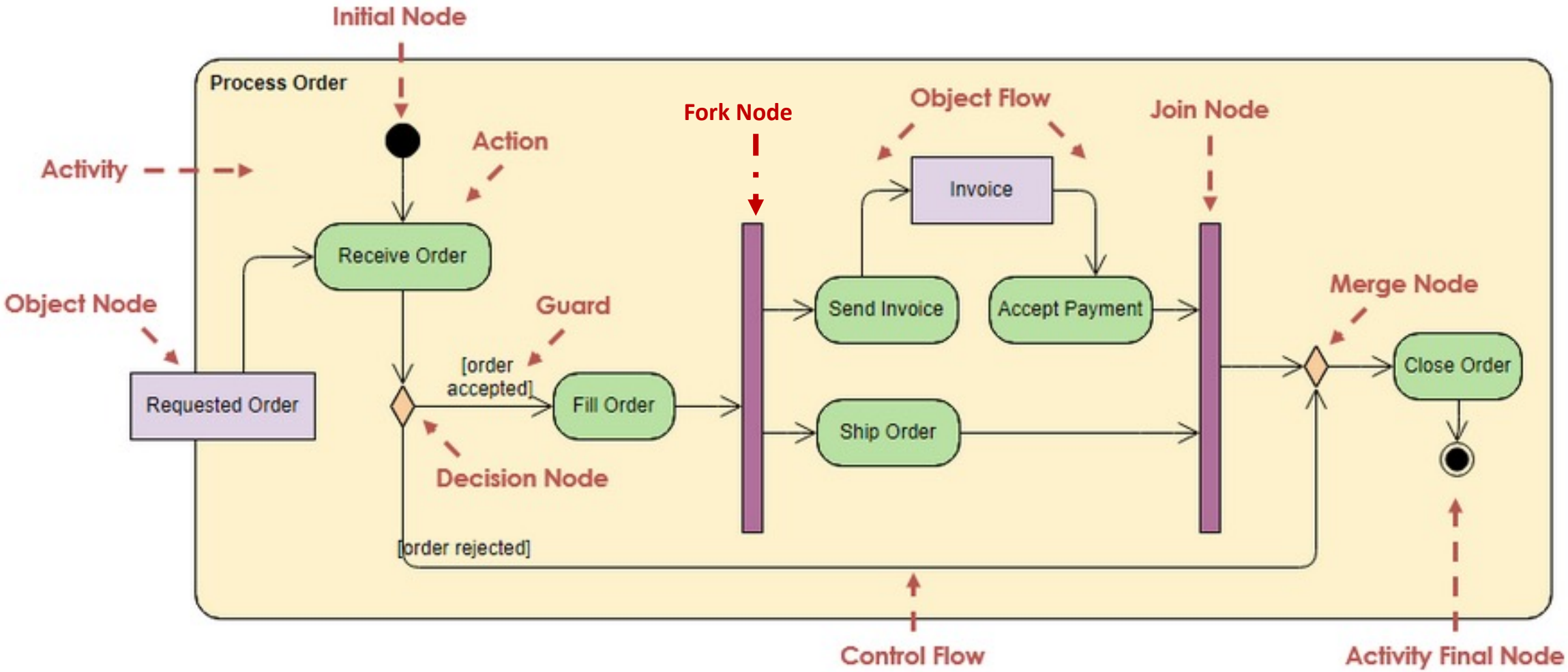
Activity Diagram

Soon Phei, Tin

What Is an Activity Diagram?

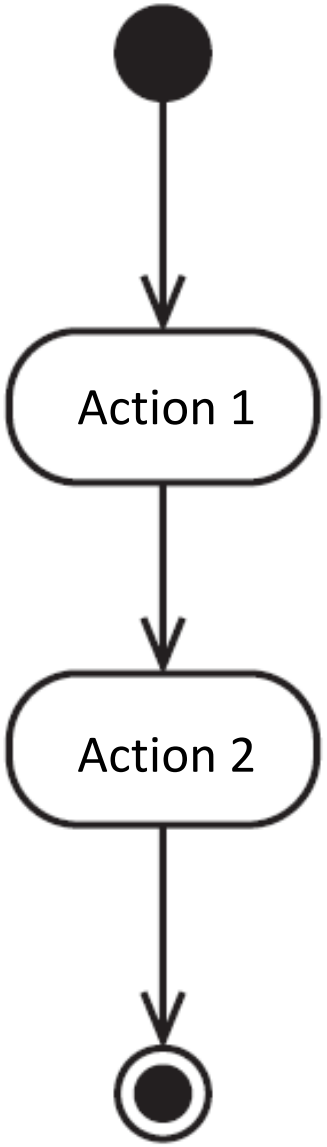
- **An activity diagram models** the flow of control through a process or workflow -- the sequence of actions, decisions, and parallel paths needed to get something done -- independent of which class or object performs each step.
- **Purpose**: capture a business process or a use case's internal logic during requirements analysis, document a workflow that spans multiple actors or systems, and spot missing decision branches or parallel work early -- before any code is written.
- Unlike a state machine diagram, which tracks how one object's state changes over its lifetime, an activity diagram tracks the steps of a process, which may span many objects and even many systems.
- **Common uses**: documenting a business workflow (e.g., order fulfillment), a use case description, or an algorithm with conditional or parallel steps.

Activity Diagram



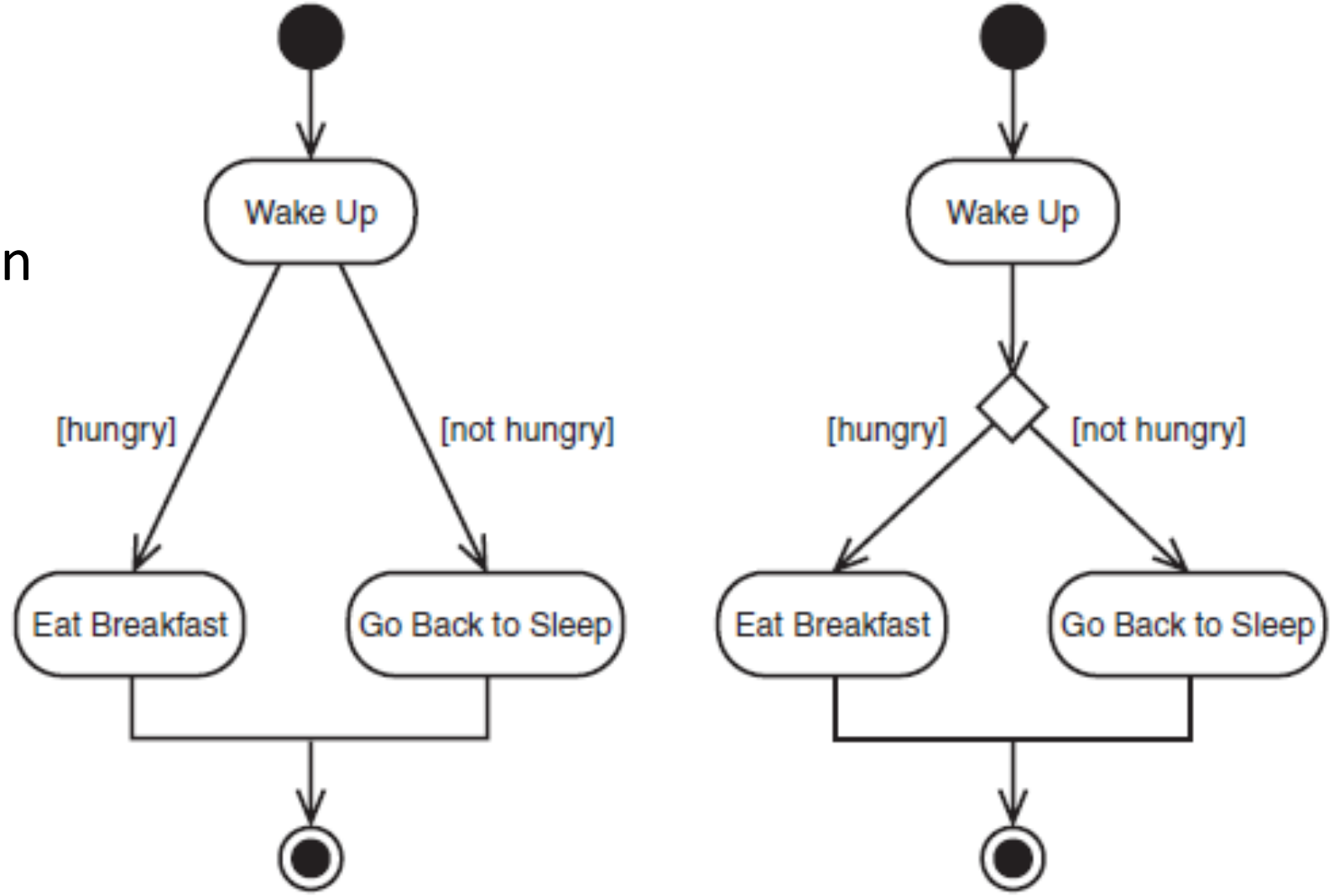
Basic

- An **action** represents an operation, a step in a business process, or an entire business process.
- Action is a named element which represents a **single atomic step within activity** i.e. that is **not further decomposed within the activity**.
- Arrow represents the transition from one activity to the next
- Filled circle represent start, bull's eye represent an endpoint



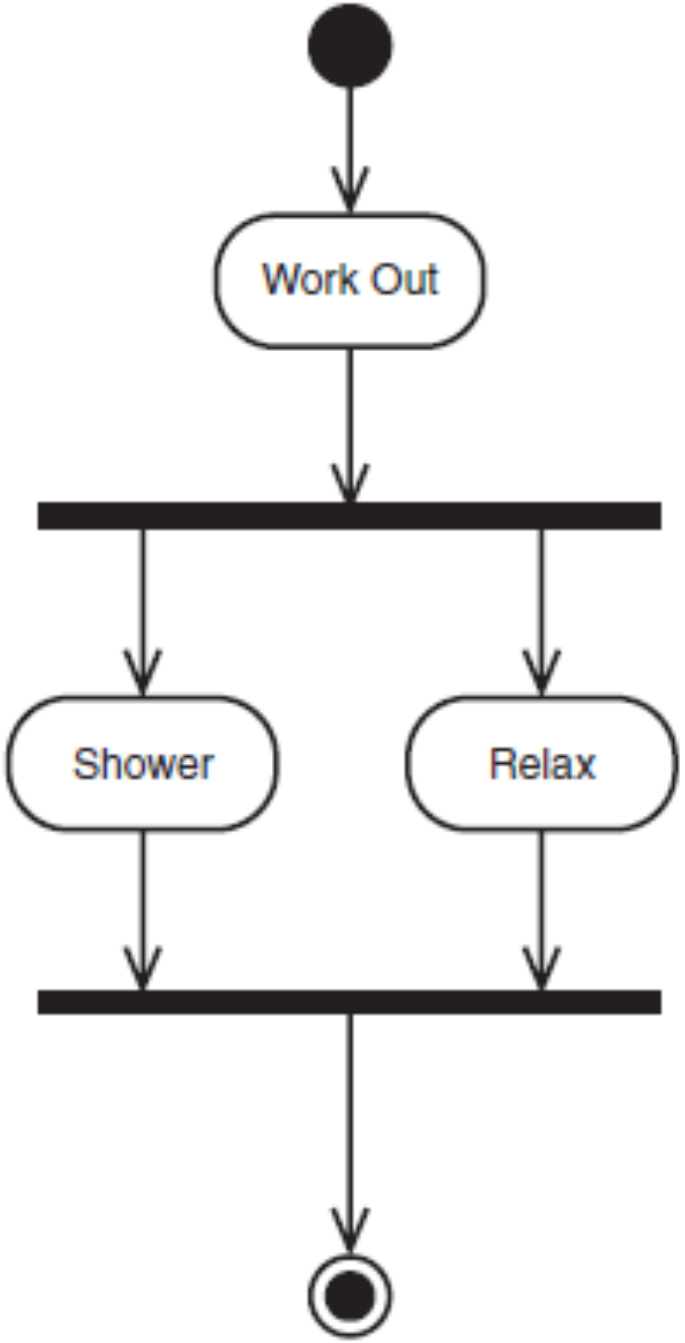
Decision

- Both are valid
- Label the condition with a squared bracket



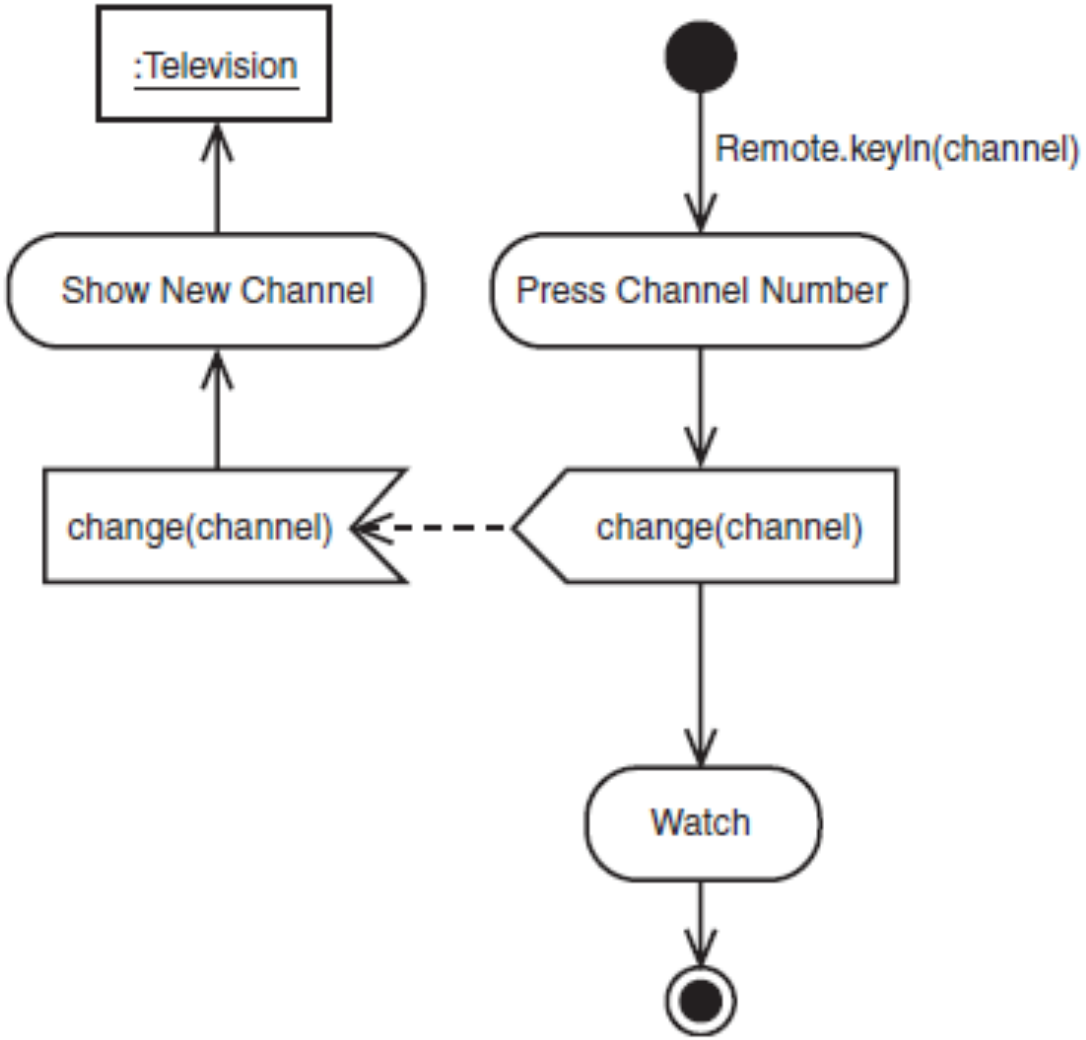
Concurrent Paths

- When two or more concurrent paths running at the same time
- Use solid bold line to split the transition arrow into multiple concurrently executing paths
- Solid bold line also used to merge the concurrent paths



Sending and Receiving Signal

- There will be situation when your process need to send or receive signal
- Convex polygon – sending signal / output event
- Concave polygon – receiving signal / input event

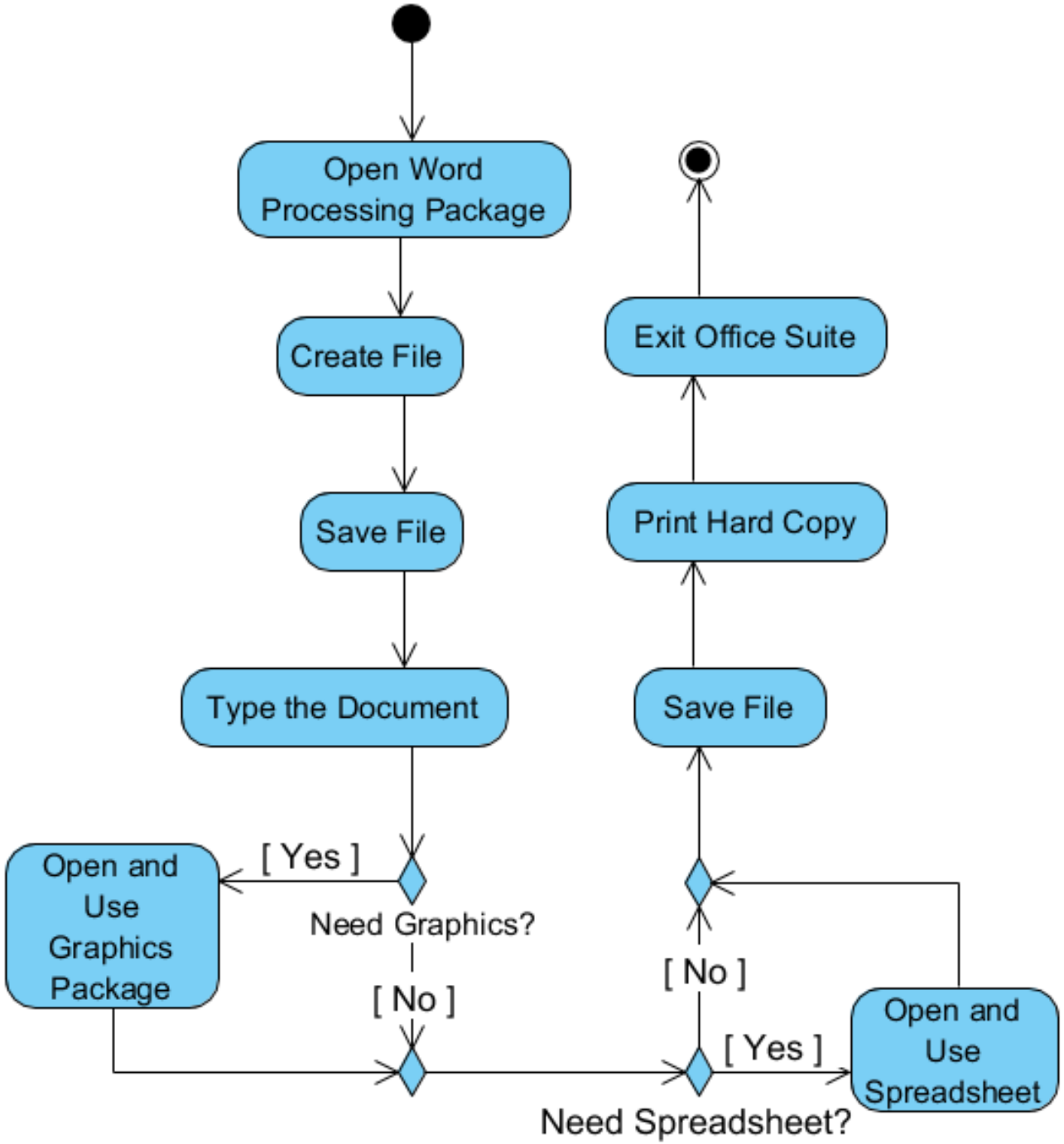


Think of the activities that go into using an office software suite to create a document. One possible sequence of activities is

1. Open the word processing package.
2. Create a file.
3. Save the file under a unique name within its directory.
4. Type the document.
5. If graphics are necessary, open the graphics package, create the graphics, and paste the graphics into the document.
6. If a spreadsheet is necessary, open the spreadsheet package, create the spreadsheet, and paste the spreadsheet into the document.
7. Save the file.
8. Print a hard copy of the document.
9. Exit the office suite.

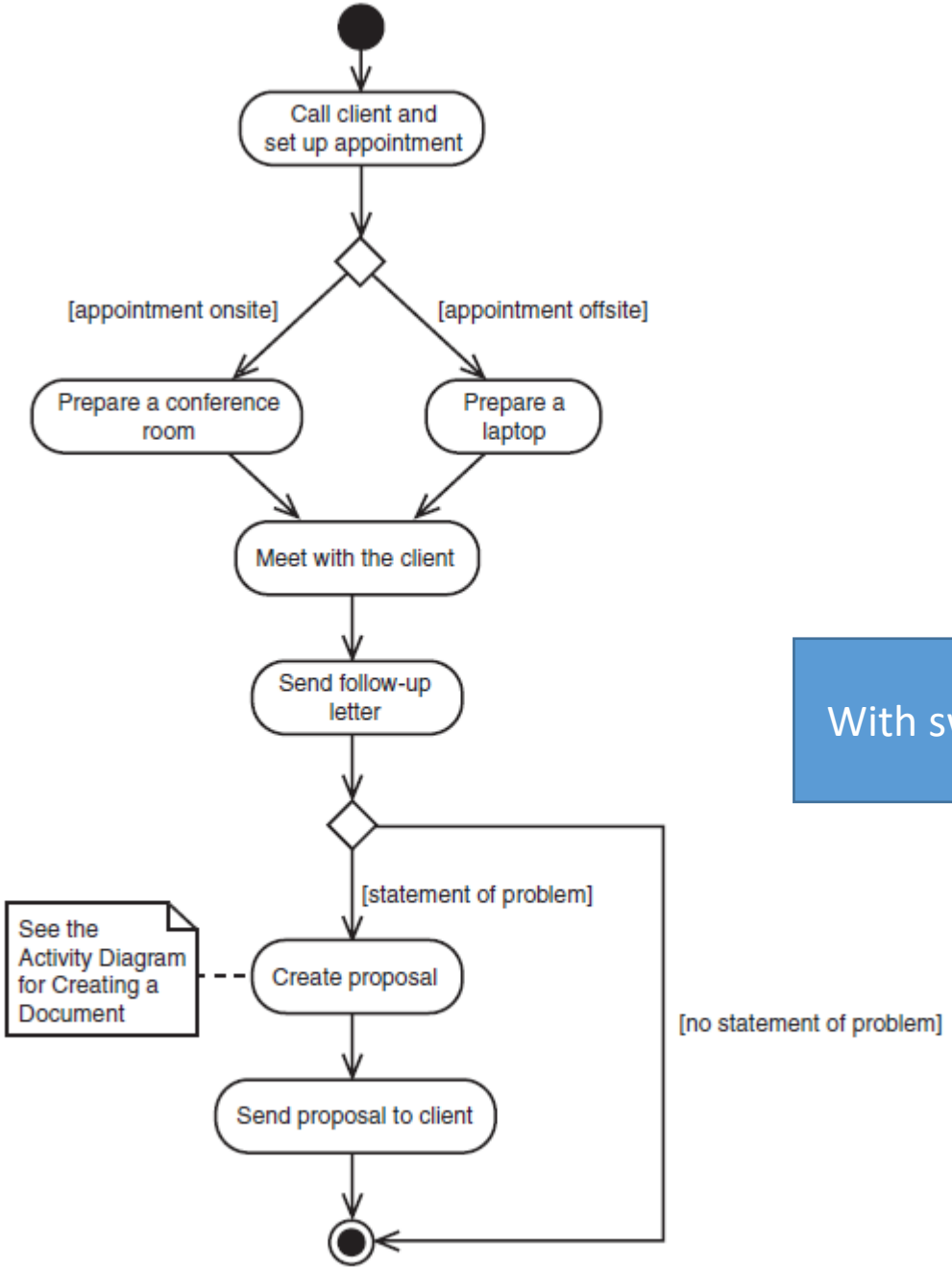
Suggested Answer

- Modify the suggested answer to include the use of Convex polygon and Concave polygon for the “Print Hard Copy” action.

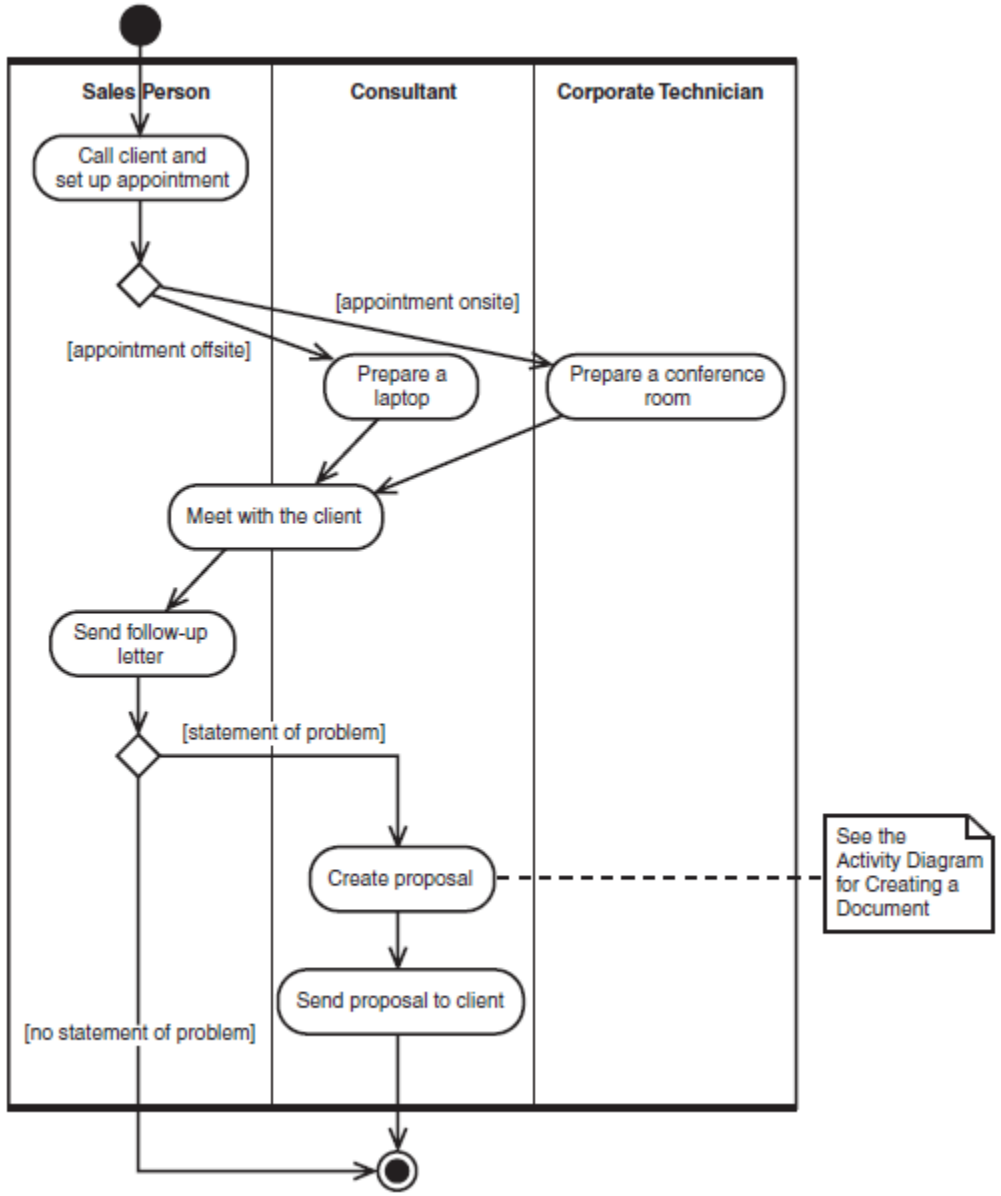


Swimlanes

- Use swimlanes to show who responsible for each activity in a process
- Separate the diagram into parallel segments called swimlanes
- Each swimlane shows the name of a role at the top
- Transitions can take place from one swimlane to another

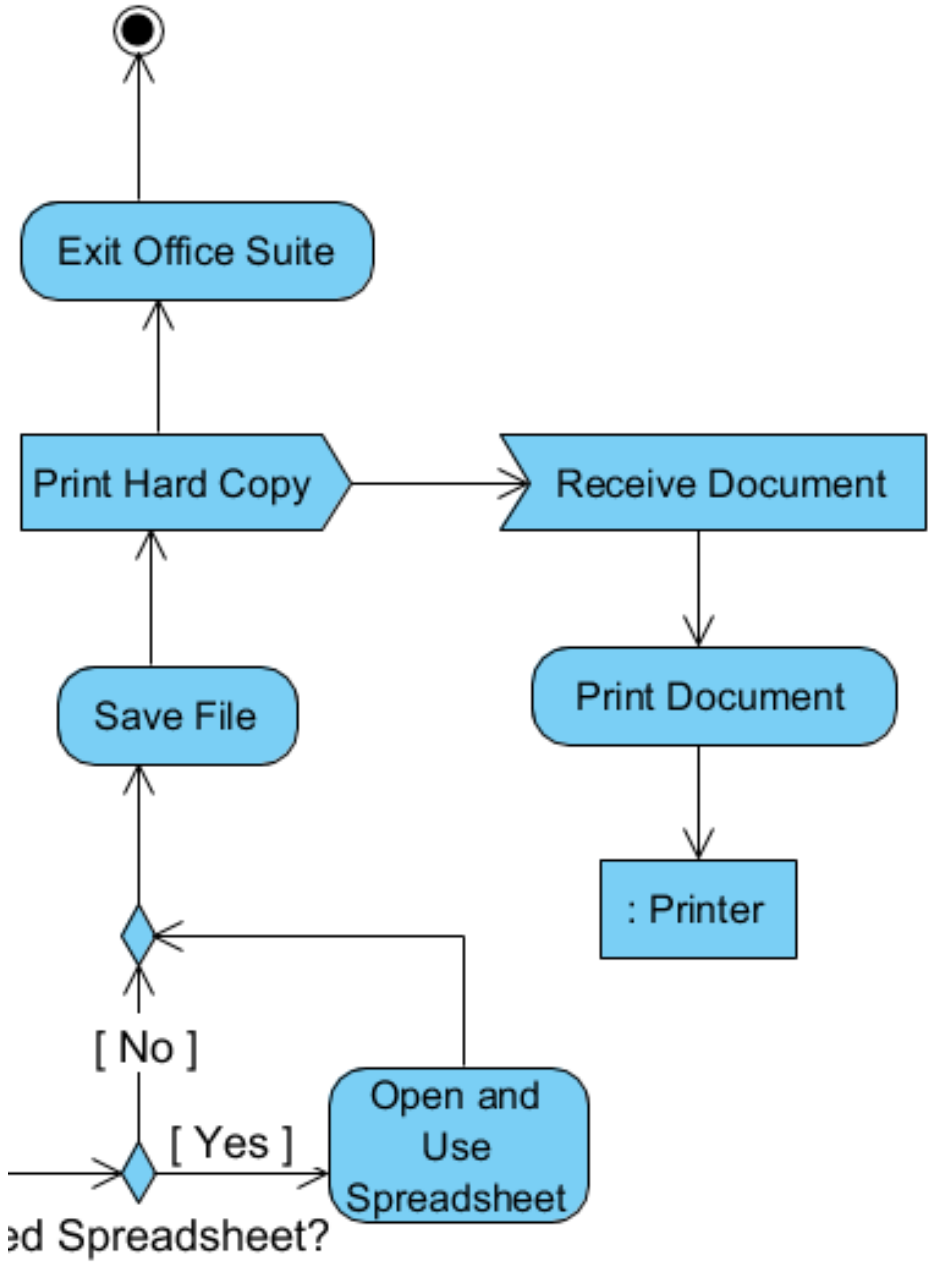


With swimlanes



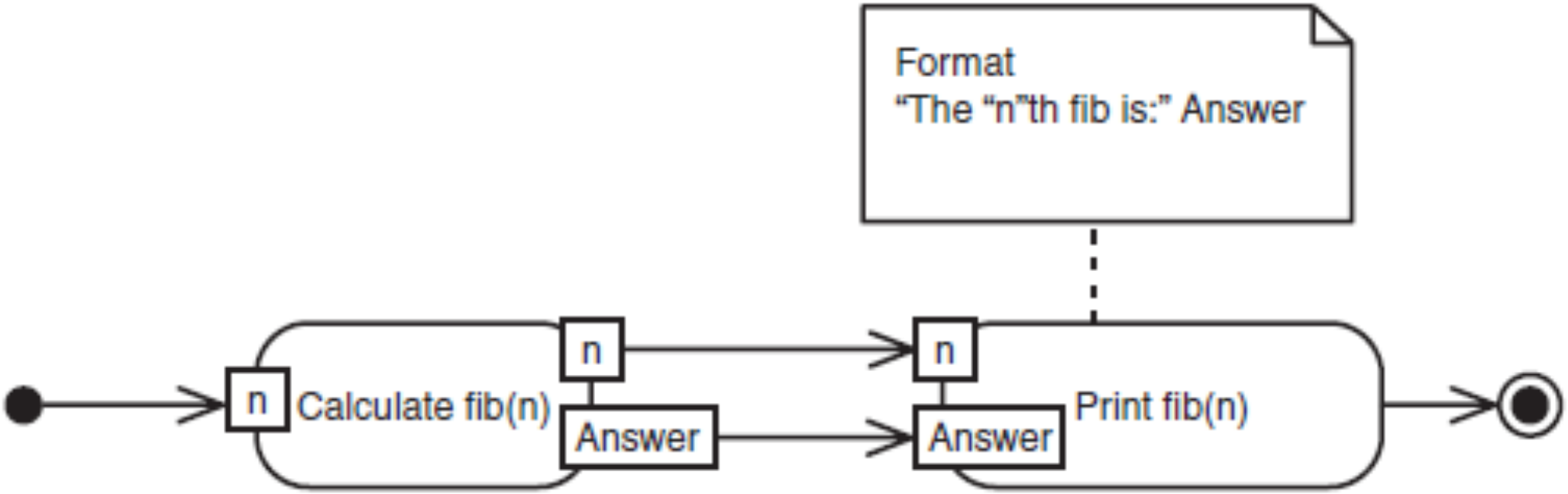
Modified Example

- Using send signal and receive signal to achieve actual printing task



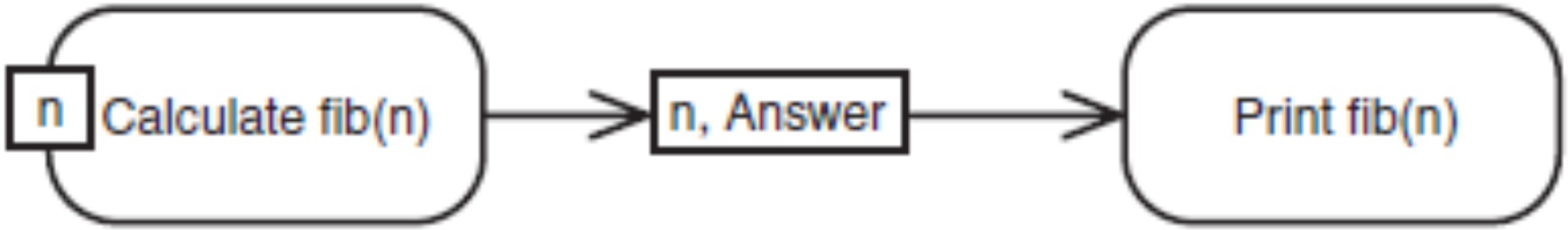
Taking parameters

- What is the problem with this notation?

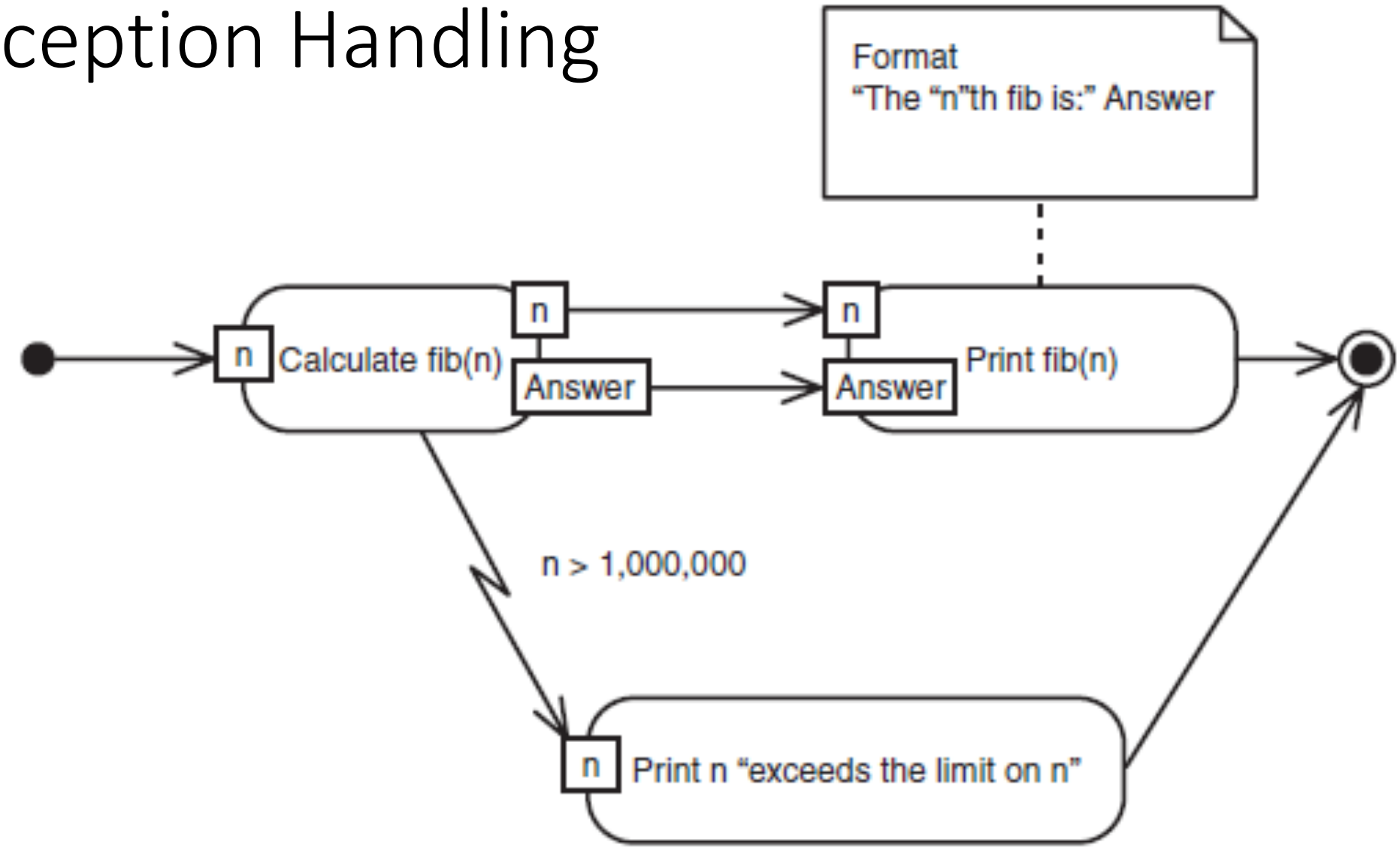


Taking Parameters

- Simplification

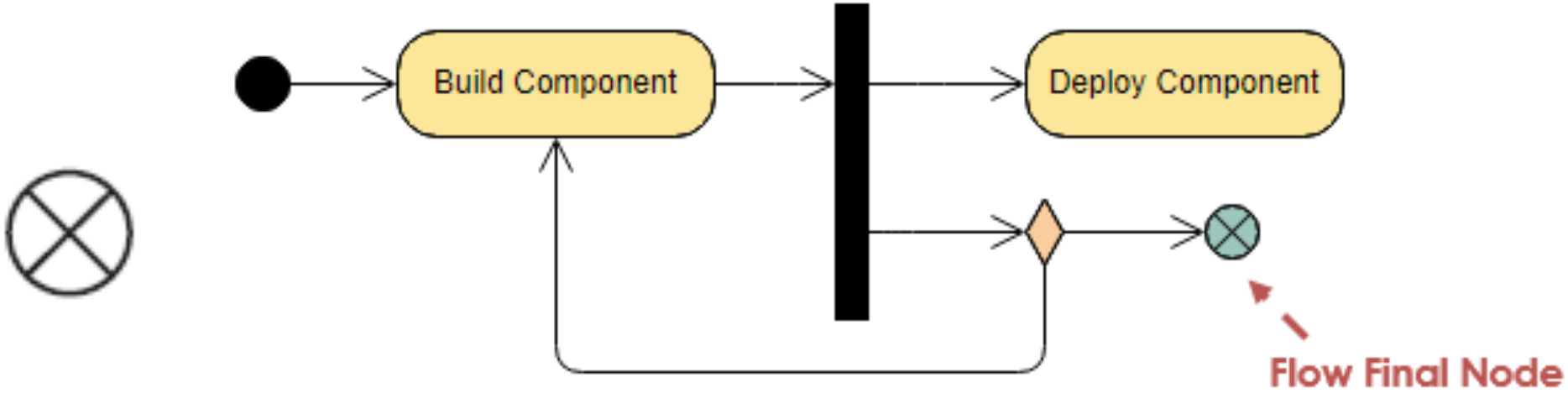


Exception Handling



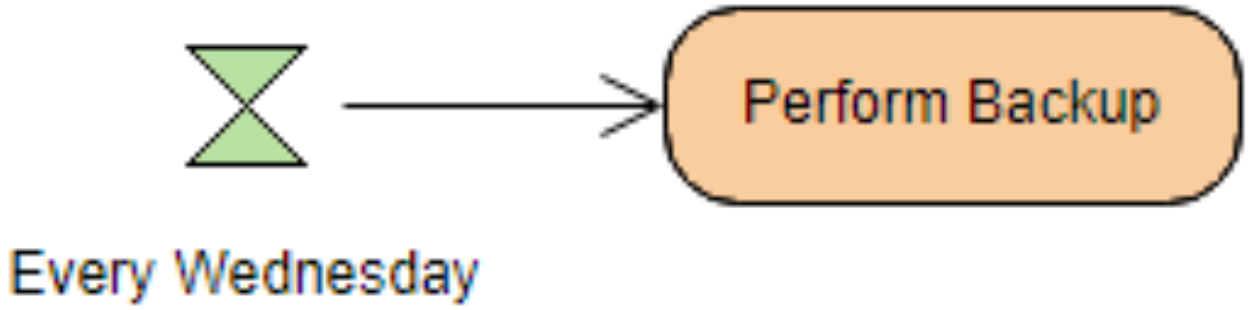
Flow Final Node

- Not Activity Final Node
- UML 2.0 has an additional control node type called Flow Final that is used as an alternative to the Activity Final node to terminate a flow.
- It is needed because in UML 2.0, when control reaches any instance of Activity Final node, the entire activity (including all flows) is terminated. The Flow Final simply terminates the flow to which it is attached.

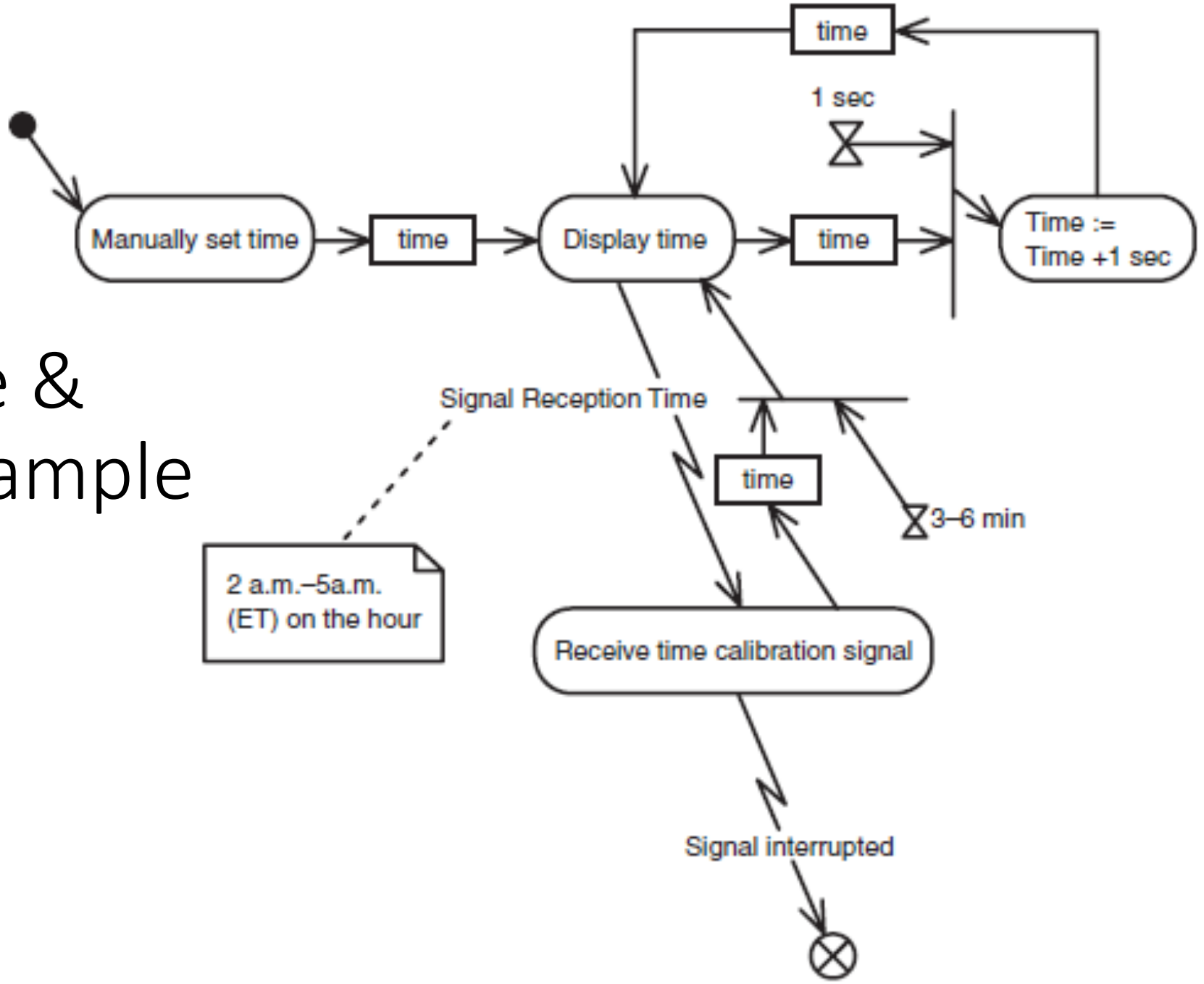


Time Event

- Time event flows when the time expression is true



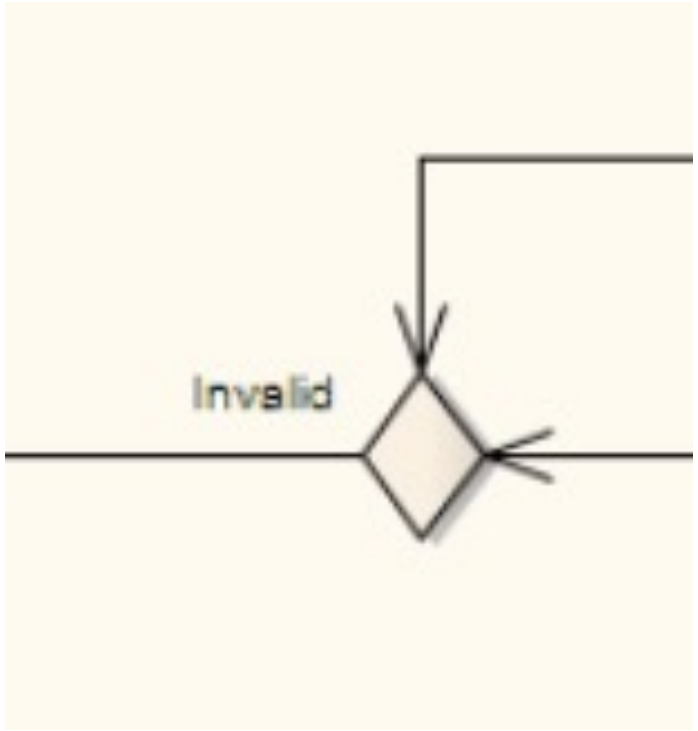
Flow Final Node & Time Event - Example



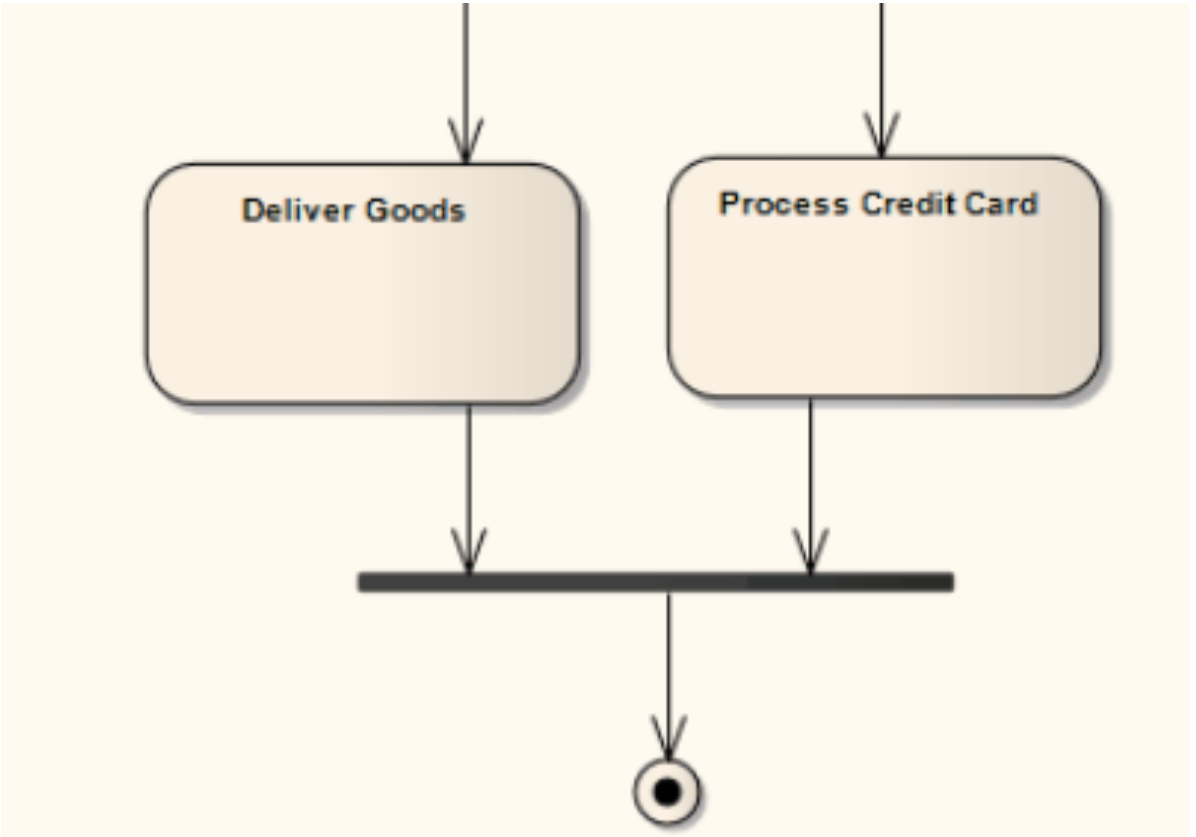
Exercises

- Making a cup of coffee
- Online shopping cart checkout process

Merge Node vs. Join Node

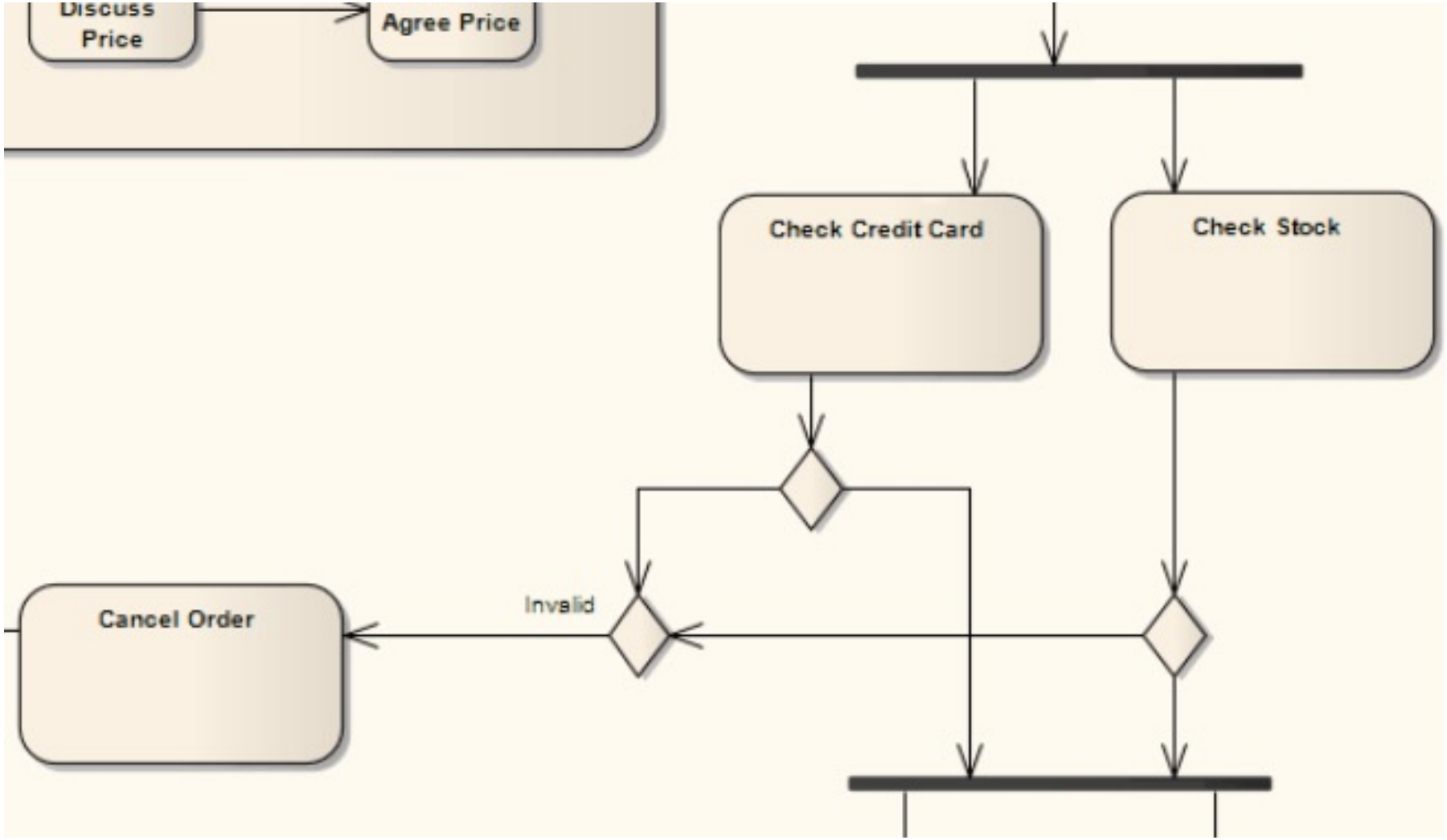


Merge Node



Join Node

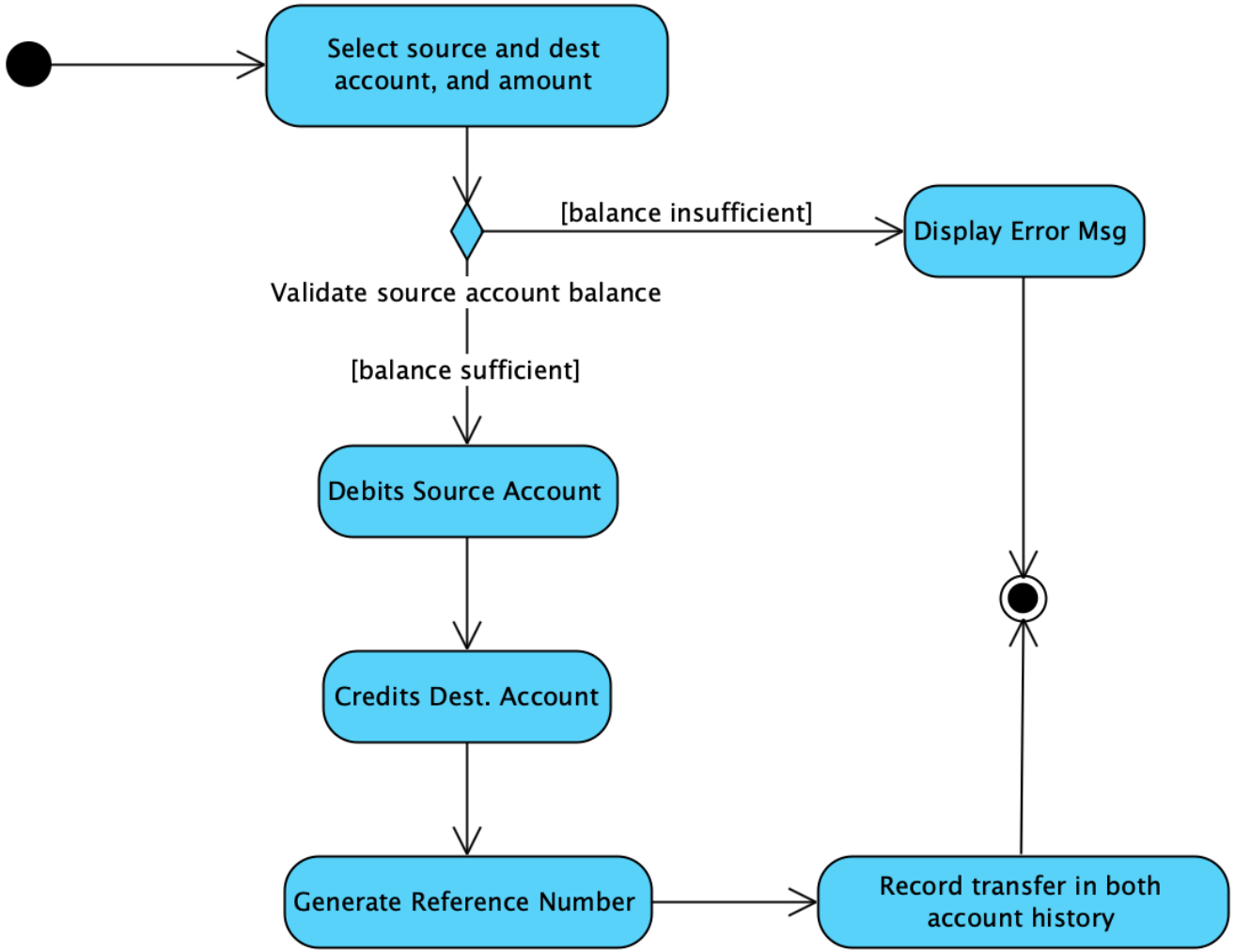
Merge Example



Online Banking Fund Transfer - Requirements

- OBS-06 — a Customer selects a source Account, a destination Account, and an amount to transfer.
- OBS-07 — the system validates the source Account and confirms it holds sufficient available balance to cover the amount.
- OBS-08 — if the balance is insufficient, the system displays an error message and the process ends here.
- OBS-09 — if the balance is sufficient, the system debits the source Account and credits the destination Account.
- OBS-10 — the system generates a unique reference number for the transfer.
- OBS-11 — the system records the transfer in both accounts' histories, and the process ends.

Online Banking Fund Transfer – Activity Diagram



Summary

- An activity diagram models the flow of control through a process: its actions, decisions and parallel paths, no matter which object performs each step
- The purpose of **swimlanes** in an activity diagramis to show which role or actor is responsible for each activity in the process
- **Merge vs. join:** a merge continues as soon as one path arrives, while a join waits for all of them
- **Activity Final vs. Flow Final:** an Activity Final ends everything, while a Flow Final ends only one flow